

Dexter Fernandes

Bristol • +44 07436007369 • dexterfernandes11@gmail.com • linkedin.com/in/dexterferns •
<https://github.com/Dexter-Fernandes>

Research Engineer

Computer Vision Engineer with 3+ years of experience delivering low-latency vision solutions for Tier 1 companies, including a construction firm in Texas and a UK oil and gas company. Proven track record of collaborating with multinational clients to deploy AI-driven vision solutions in fast-paced dynamic environments. Experienced in applying state-of-the-art AI research while leading testing, QA, and monitoring processes.

WORK EXPERIENCE

Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation.
- Orchestrated a sensor-data cleansing pipeline for 3.6 million records using Pandas and Apache Airflow, reducing model rollout time from 3 days to under 24 hours.
- Built multimodal data ingestion, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, processing 3.6 million records and reducing model rollout time from 3 days to under 24 hours.
- Refactored legacy services into Dockerised microservices with Kafka-based messaging, structured logging, health checks, retries, and safe restarts, improving fault isolation, increasing MTBF by 35%, and reducing live video pipeline crashes by 67%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, improving maintainability and reducing defects.
- Documented deployment guardrails, failure modes, and recovery procedures, improving maintainability for engineers supporting fielded systems.
- Implemented Git and Jenkins-based continuous integration and validation workflows for edge inference modules for 24/7 deployments to improve release consistency across hardware targets.
- Collaborated with cross-functional engineering and client teams to translate performance metrics into actionable deployment and optimisation decisions.
- Integrated AI modules with production systems in collaboration with other teams to deliver fault-tolerant edge deployments.
- Engineered production-grade AI software for real-time perception and safety analytics, delivering deterministic performance across 24/7 oil rig operations.
- Validated deterministic model performance under extreme field conditions at a British Petroleum site, ensuring reliability across 24/7 operations.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.
- Led on-site validation and diagnostics on BP oil-rig deployments, troubleshooting model failures under environmental stressors and translating performance findings into fixes that improved robustness in live operation.
- Built synthetic image generation and augmentation procedures with OpenCV and scikit-image to improve model robustness under harsh industrial conditions, reducing false positives and supporting long-term operational reliability.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.

- Diagnosed and resolved production failures through on-site analysis and structured validation checklists, reducing crash frequency by 67% in live video pipelines.
- Built a Django-based internal computer vision application with a web dashboard to visualise inference, system health, and performance metrics, enabling faster debugging.
- Designed adversarial training experiments (occlusion, lighting variation, motion artefacts), benchmarking robustness via F1 and failure-rate metrics, achieving a 15% F1-score improvement within 3 months.
- Developed a repeatable optimisation workflow (baseline to FP16 to validation) with defined acceptance thresholds, enabling consistent latency, memory, and stability outcomes across deployments.
- Built benchmarking and profiling harnesses to evaluate open-source models (YOLOv5, EfficientDet) and quantify accuracy, latency, and stability pre- and post-optimisation for regression detection.
- Diagnosed and resolved production issues in Dockerised microservices, improving service continuity through stronger logging, health checks, safe restarts, and workload isolation.
- Implemented reliability patterns and operational observability, using timeouts, retries, message validation, and structured logs and metrics to improve root-cause analysis and maintain stable behaviour under load.
- Added operational observability using structured logs and metrics, enabling faster root-cause analysis and documented failure modes and recovery procedures to reduce operational overhead.
- Added operational observability using structured logs and metrics, enabling faster root-cause analysis and documented failure modes and recovery procedures to reduce operational overhead.
- Developed an internal web dashboard in Django to surface inference outputs, system health, and performance metrics, enabling faster debugging and root-cause analysis across deployed systems.
- Led validation, benchmarking, and observability for fielded AI deployments, reducing crash frequency by 67% and documenting failure modes, recovery procedures, and deployment guardrails for engineers operating live systems.
- Applied image preprocessing and artefact-detection workflows with OpenCV and scikit-image to identify mud and sand contamination on camera feeds, reducing false positives by 13% in harsh industrial environments.
- Benchmarked and optimised edge inference workflows across latency, memory, and stability constraints, defining repeatable FP16 validation paths and acceptance thresholds for reliable deployment across hardware targets.
- Led field validation on BP oil-rig deployments, diagnosing model failures under lighting, motion and environmental stressors and translating findings into practical fixes for live operation.
- Applied statistical and experimental methods including Bayesian optimisation, ablation studies, hypothesis testing, and confidence intervals to support model selection, threshold tuning, and failure analysis in production.
- Worked with multinational stakeholders, from project managers to drill rig operators, translating operational problems into deployable data science solutions and communicating findings to both technical and non-technical stakeholders.
- Worked with cross-functional engineering and client teams to translate performance findings into deployment and optimisation decisions for production systems.
- Owned production monitoring and post-deployment validation for deployed vision systems, using observability tooling, failure analysis, and drift tracking to diagnose issues early and improve reliability in live environments.
- Applied image preprocessing and artefact-detection workflows with OpenCV and scikit-image to identify camera-feed contamination, reducing false positives by 13% in production vision pipelines.
- Improved production video inference reliability by adding structured logging, health checks, retries, and safe restart behaviour, reducing live pipeline crashes by 67% in 24/7 operation.
- Debugged training dynamics and convergence issues in production CV models, balancing model complexity, augmentation, and validation strategy to improve generalisation under lighting variation, occlusion, motion artefacts, and camera contamination.
- Hardened deployed vision systems with Dockerised microservices, Kafka messaging, structured logging, health checks, retries, and safe restarts, reducing live video pipeline crashes by 67% and improving root-cause analysis.
- Analysed model performance and failure patterns from live video deployments, turning F1, false-positive and stability metrics into deployment guardrails and optimisation recommendations for engineering and operations teams.

Lead Software Engineer - AI • Full-time

- Led a team to design, train, and deploy production object detection and tracking models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints.
- Built low-latency perception pipelines using OpenVINO, and C++17, tripling throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel VPU hardware under tight power and memory constraints.
- Deployed quantised vision models using FP16/INT8 optimisation, pruning and layer fusion, reducing model size by 50% while preserving production accuracy targets.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using OpenCV and Albumentations.
- Conducted systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Led a 4-engineer team to design, train, and deploy CNN-based object detection and tracking models for industrial safety applications, reducing false positives by 33% and achieving sub-50 ms inference latency on edge devices.
- Optimised neural network architectures using quantisation-aware training, structured pruning, and layer fusion reducing model size by 50% with Keras and Intel OpenVINO.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Collaborated with hardware, software, and QA teams to integrate CV models into deployed systems, and presented performance findings to leadership and clients to support rollout, optimisation, and reliability decisions.
- Engineered production-grade edge perception software for safety-critical environments, delivering deterministic real-time behaviour on Intel VPU devices.
- Mentored junior engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.
- Presented model performance, failure analysis and deployment trade-offs to engineering, QA, hardware and client teams; findings directly shaped rollout and optimisation decisions.
- Built and maintained large-scale image and sensor datasets, automating labelling, versioning, and performance tracking to support anomaly investigation and continuous improvement cycles.
- Conducted optimisation experiments across quantisation-aware training, pruning, layer fusion, FP16/INT8 deployment and profiling, reducing model size by 50% while preserving accuracy under edge constraints.
- Implemented optical flow-based ego-motion estimation feeding motion representations into CNNs to classify vehicle state in production pipelines.
- Built asynchronous inference pipelines using OpenVINO and C++17 to maximise throughput on Intel VPU edge hardware under real-time latency and power constraints.
- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency, and operational cost while maintaining safety-critical performance, supporting production cloud-hosted ML workloads on Microsoft Azure.
- Designed data handling and retention practices aligned with GDPR principles, including data minimisation and controlled access, within production computer vision systems.
- Owned end-to-end production ML pipelines spanning data ingestion, labelling, training, optimisation, deployment, monitoring, and field validation for industrial safety systems, improving deployment reliability and release consistency.
- Provided technical leadership to a 4-member engineering team, setting experimentation standards, reviewing model performance, and guiding optimisation strategies for production deployments.
- Implemented an asynchronous inference pipeline in C++ with OpenVINO, overlapping pre-processing, inference, and post-processing to reduce end-to-end latency and improve pipeline utilisation on Intel GPU targets.
- Profiled and optimised C++ inference and pre and post-processing hotspots using Intel VTune Profiler for targeted performance fixes in our CNN inference pipeline.

- Integrated Boost C++ libraries into production pipelines, resolving dependency and build issues and maintaining reproducible deployments.
- Defined optimisation workflows spanning quantisation-aware training, pruning, layer fusion, FP16/INT8 deployment, and profiling, reducing model size by 50% while meeting accuracy, latency, and memory targets.
- Built automated data, labelling, and performance-tracking pipelines, repurposing older models for annotation and cutting manual labelling effort by 80% and also developed synthetic-data tooling to improve edge-case coverage and reduce manual labelling time by 60%.
- Worked closely with hardware, software, and QA teams to integrate perception modules into deployed systems and support rollout, optimisation, and reliability decisions.
- Optimised model architectures using quantisation-aware training, structured pruning, layer fusion, FP16/INT8 deployment, and targeted profiling, reducing model size by 50% while meeting accuracy, latency, and memory constraints.
- Owned end-to-end production ML pipelines from data ingestion and labelling through training, optimisation, deployment, monitoring, and field validation for industrial safety systems.
- Built a low-latency, high-reliability computer vision PoC for a zinc ore refining plant in South Africa, and developed a user-facing hazard detection application for an Oman-based client.
- Built and maintained a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising camera streams and running CV-based incident detection across both feeds under edge compute constraints.
- Implemented optical flow-based ego-motion estimation to feed motion representations into CNNs for vehicle-state classification in production pipelines.
- Explored semantic and instance segmentation models, including DeepLabv3+, Mask R-CNN, U-Net-style architectures, and SegFormer, to support annotation review, foreground/background separation, and failure analysis for industrial camera feeds.
- Built YOLOv5-based visual quality inspection for John Deere engine paintjobs on a conveyor line, detecting paint defects, contamination, and colour inconsistency as part of a manufacturing anomaly detection workflow.
- Used confusion-matrix-driven failure clustering to diagnose recurring CV failure modes, prioritise dataset fixes, and improve model robustness across camera feeds.
- Integrated perception modules with hardware, software and QA teams, translating model performance, latency and failure analysis into rollout decisions for deployed safety systems.

PROJECTS

FinanceBench RAG Evaluation Harness

Jun 2026

- Built a full-pipeline eval harness for FinanceBench-style financial QA, covering dataset validation, evidence-page matching, LLM baseline eval, dense retrieval, RAG generation, and automated scoring with structured outputs per run.
- Defined evidence-level retrieval metrics (`doc_hit@k`, `page_hit@k`, `evidence_text_hit@k`, `MRR@k`, numeric match, evidence overlap) to verify answers were grounded in the right source pages, not just plausible.
- Ran three controlled eval modes (closed book, oracle context, dense RAG) with MockLLM tests for CI, optional Ollama for local inference, and YAML configs, JSONL predictions, and Markdown reports for reproducibility.
- Designed the harness to extend cleanly: embedding swaps, hybrid retrieval, reranking, citation checks, hallucination labels, and failure analysis can all be added without reworking the core loop.

Modern Edge CV Benchmarking

Mar 2026

- Benchmarked modern computer vision models including YOLOv8, YOLOv10, RT-DETR, EfficientViT, Grounding DINO, MobileSAM, and Depth Anything v2 to compare accuracy, latency, model size, and deployment suitability.
- Evaluated Vision Transformer and foundation-model approaches for object detection, segmentation, depth estimation, and open-vocabulary visual understanding, with a focus on production-style CV trade-offs.
- Exported models to ONNX and evaluated OpenVINO deployment paths, strengthening practical understanding of model optimisation, inference acceleration, and edge deployment constraints.
- Built benchmark comparisons using mAP, inference time, and model-size measurements to support model selection for low-latency visual AI systems.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on the Clearpath Husky platform, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios using ROS2 and rosbag2-based workflows to assess mapping and localisation behaviour.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL, and developed ROS2 nodes with tf2, message synchronisation, and sensor data playback workflows to maintain frame consistency and debug transform timing issues.
- Worked with camera and sensor ingestion in ROS2 pipelines, including timestamp alignment and downstream processing for SLAM and perception tasks.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

CERTIFICATIONS

Building RAG Agents with LLMs

Nvidia

Jun 2026

Deep Learning Specialisation

deeplearning.ai

Jul 2019

SKILLS

- **SLAM, Camera and Data Pipelines:** Camera ingestion, Cartographer, Ceres, Eigen, Filtering, GLIM, LiDAR SLAM, LIO-SAM, Multi-camera perception, Pose graph optimisation, Registration, RTAB-Map, Scan matching, Segmentation, Video and sensor data processing pipelines, Visual data streams
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Health checks, Intel VPUs, Microsoft Azure, MLFlow, Post-deployment validation, Production monitoring, Structured logging
- **Programming & Tooling:** Bash, C++17, CMake, colcon, Django, FastAPI, Git, Intel VTune, Linux, NumPy, OpenCV, Pandas, PG vector, PIL, PostgreSQL, Python, Qt, Scikit-image, Scikit-Learn
- **Deep Learning and Optimisation:** Bayesian Hyperparameter Optimisation, Convolutional Neural Networks, FP16/INT8 deployment, Layer fusion, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, YOLO
- **Edge AI and Inference:** Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU, NVIDIA Jetson AGX Xavier
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **Embedded and Low-level:** 8051 Assembly, ADC/DAC interfacing, Arduino C, Embedded C, Interrupt-driven I/O, Timers and counters, UART serial
- **Point Cloud & Vision Processing:** Denoising, Open3D, OpenCV, Point Cloud Library, Scikit-Image

- **Robotics and Middleware:** Closed-loop robot control, Gazebo Harmonic, Inverse kinematics, Nav2, rclcpp, rclpy, Real-time perception systems, robot_localization, ROS2 (Robot Operating System 2), rosbag2, sensor data playback, Sensor integration, tf2, timestamp synchronisation
- **Large Language Models:** Citation-grounded response generation, Error analysis, Experimentation and eval, Information extraction, Model fine-tuning, Retrieval-augmented generation (RAG), Search and Retrieval, Workflow automation
- **Computer Vision & Image Processing:** Anomaly detection, Data augmentation, Defect detection, Image processing, LiDAR extrinsics calibration, Object detection, Object tracking, Segmentation, Sensor calibration, Visual quality inspection
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Prompt engineering
- **Computer Vision Models:** ByteTrack, DeepLabv3+, Depth Anything v2, EfficientDet, EfficientViT, Faster R-CNN, Grounding DINO, Mask R-CNN, MobileSAM, RT-DETR, SegFormer, SSD MobileNet V2, U-Net, YOLOv10, YOLOv5, YOLOv8
- **Datasets, Annotation & Evaluation:** Automated labelling, Cityscapes, COCO, Confusion-matrix-driven failure clustering, CVAT, Dataset versioning, DVC, F1-score, ImageNet, IoU, KITTI, Latency benchmarking, mAP, Model-size benchmarking, PASCAL VOC, Precision/Recall, Synthetic data generation